Homework 9: Plastics

Paper track. Pairs with Home Lab 9.

Part A: Density ladder (3 pts each)

Reference densities (g/cm^3): isopropyl 0.79 · veg oil 0.92 · water 1.00 · saturated salt water 1.20 · corn syrup 1.33.

For each result, give the density **bracket** and name the plastic (use the burn result if given):

1. Floats in veg oil, sinks in isopropyl.
2. Floats in water, sinks in veg oil. Squeeze test: soft and stretchy and fairly clear.
3. Sinks in water, floats in saturated salt water. Burn: dense black sooty smoke, styrene smell.
4. Sinks in saturated salt water, floats in corn syrup. Burn: self-extinguishes, acrid (HCl) smell.
5. Sinks in saturated salt water, floats in corn syrup. Burn: from a drink bottle, melts with black smoke, partly self-extinguishing.

Part B: Resin codes & uses (1 pt each)

Match the code to the plastic and give one common use: 1, 2, 4, 5, 6.

Part C: Types & history (2 pts each)

1. Explain the difference between a thermoplastic and a thermoset.
2. All rules plastics are thermo______.
3. First fully synthetic plastic — name, inventor, year.
4. Roughly what % of plastic gets recycled?
5. How do you tell LDPE from HDPE by feel?

Part D: Scenario (3 pts)

A shard of clear, very hard plastic is found at a scene where a window-like panel was smashed. It sinks in saturated salt water, floats in corn syrup, and the burn card reads "faint sweet smell, very slow to burn, sooty." One suspect installs bullet-resistant glazing; one sells acrylic display cases. Which plastic is it, and which suspect does it implicate?